



Plastering, Pointing & Painting

Arbaz Mubarak Kazi, B.E, M.E (Civil Engineering)

Email: arbaz.kazi@vcet.edu.in

Website: <https://sites.google.com/vcet.edu.in/arbazkazi>



Plastering - Definition

- ❖ Plastering is the process of applying mortar coats to various surfaces, including walls, columns and ceilings, to achieve a smooth and finished appearance.
- ❖ The mortar used for plastering can be composed of different materials, including lime mortar, cement mortar or a combination of both, known as lime-cement mortar.
- ❖ Lime mortar typically has a fat-lime-to-sand ratio of 1:3 or 1:4. When hydraulic lime is used, the mix proportion becomes 1:2. Cement mortar, with a mix ratio of 1:4 or 1:6, is a commonly employed choice for plastering, with richer mixes reserved for exterior walls.
- ❖ For a cost-effective yet high-quality solution, some opt for lime-cement mortar with proportions such as 1:1:6, 1:1:8 or 1:2:8, blending the advantages of lime mortar and cement mortar.



Objectives of Plastering

- ❖ Concealing any flaws or defects in the underlying construction work.
- ❖ Providing a sleek surface that resists dust accumulation.
- ❖ Enhancing the visual appearance of the surface.
- ❖ Shielding the wall from environmental elements, particularly rainwater.
- ❖ Safeguarding against damage from pests and vermin.



Requirements for Ideal Plaster

An ideal plaster should possess certain qualities to ensure durability, strength and a good finish. These properties help the plaster perform effectively under different environmental conditions. The requirements of Ideal Plaster are:

- **Adhesion:** Ensuring the plaster adheres effectively to the substrate.
- **Durability:** Creating a sturdy and long-lasting plaster that can withstand wear and tear.
- **Moisture Barrier:** Forming a protective barrier that prevents moisture penetration.
- **Cost-Effectiveness:** Achieving these objectives in an economically efficient manner.

WORK PROCESS OF PLASTERING





Defects in Plastering

Blistering of Plastered Surface

Blistering is the formation of small patches of plaster, swelling out beyond the plastered surface, arising due to late slaking (addition of water to lime) of lime particles in the plaster. This defect is usually caused due to the uneven mixing of plaster.



Plaster De-bonding

De-bonding occurs when a plaster is separated from the wall. It can be caused by an excessively thick plaster layer, inadequate substrate preparation or may be due to a dusty, oily or dry substrate.



Defects in Plastering

- ❖ **Crazing** – It is a network of fine cracks like spider web. They are usually very fine and do not extend through the whole depth of the plaster. It occurs due to presence of excess fine content in the sand or due to dry base on which plaster is applied – when base absorbs the water and fines accumulate on the surface, it leads to crazing.
- ❖ **Separation crack at joints** – It usually occurs at joints of two different materials for example at junction of RCC & Brick work. It occurs due to differential thermal movement.
- ❖ **Crack with Hollowness** – This crack occurs due to hollowness in plaster. Other reasons could be extra water in the plaster mix or due to poor workmanship.





Defects in Plastering

Efflorescence on Plastered Surface

When a newly constructed wall dries out, the soluble salts are brought to the surface, and they appear in the form of a whitish crystalline substance. This is called efflorescence. Efflorescence is formed on plasters when soluble salts are present in plaster making materials as well as building materials such as bricks, sand, cement etc. Even water used in the construction work may contain soluble salts. It seriously affects the adhesion of paint with the wall surface and causes further problems.





Defects in Plastering

Falling Out of Plaster

This defect can happen in two forms – Flaking of plaster and peeling off plaster.

Flaking of plaster: The formation of a small loose mass on the plastered surface is known as flaking. It is mainly due to bond failure between successive coats of plaster.

Peeling off plaster: The plaster from some portion of the surface comes off and a patch is formed. This is termed as peeling. It is also mainly due to bond failure between successive coats of plaster.





Defects in Plastering

Popping of Plaster

Popping is the formations of conical like holes that break out of the plaster. It is caused due to the presence of contaminant particles such as burnt lime or other organic materials in the mix of mortar.

Loose Plaster

When the plaster gets displaced on external impacts like application of material or tapping, etc, it is termed as loose plaster. This is caused mainly due to improper mixture and inadequate curing.



Pointing - Definition

The process of raking out mortar joints in brick or stone masonry to a depth of 10 mm to 20 mm and filling them with a fresh, richer mortar mix. It protects exposed joints from weather and improves the look of the wall. Following below are the purpose of pointing

- ❖ **Weather protection:** Seals gaps to stop rain, dampness, and wind from damaging the wall.
- ❖ **Better look:** Highlights the neat patterns and colors of exposed bricks or stones.
- ❖ **Higher strength:** Fixes weak or old joints to make the masonry stable.
- ❖ **Cost savings:** Replaces full plastering when you want to show off the natural texture of high-quality brick or stone work.



Defects in Plastering

After knowing the defects, read some precautions to be taken while plastering:

- ❖ The areas that need to be plastered should be washed and kept wet.
- ❖ All loose material sticking to the masonry should be removed before plastering.
- ❖ The joints should be cleaned by brushing or scrapping with wire brushes.
- ❖ The uniform level is tested with plumb-bob before & after applying the plaster.
- ❖ Specified cement, sand ratio should be used while making cement mortar.
- ❖ Ensure proper bonding between the surface of masonry and plaster.



Types of Pointing

- ❖ **Flush Pointing:** In flush pointing, mortar is filled into the joint, pressed, and finished flush with the brick or stone edges, creating a smooth appearance. The final touch involves trimming the edges neatly with a trowel or straight edge. While it may lack visual appeal, it's durable and prevents dust and water accumulation, making it a common choice in construction.
- ❖ **Weathered Pointing:** In weathered pointing, fresh mortar is pressed into the joint, and the top of the horizontal joints is neatly pressed back by 3-6 mm with a pointing tool, resulting in a sloping appearance from the top to the bottom of the joint.
- ❖ **Keyed Pointing:** Keyed pointing involves pressing mortar into joints with a trowel and finishing it with the masonry's face. Small-diameter steel is used to press the joint back lengthwise, creating a curved arc groove in the mortar joint, offering an aesthetically pleasing finish. It's commonly used for superior-quality work, especially for vertical joints.



Types of Pointing

V-Grooved Pointing (V Pointing)

Similar to keyed pointing, V pointing entails filling and pressing mortar into a joint, followed by the creation of a V-shaped groove in the joint using a V-shaped tool.

Beaded Pointing

Beaded pointing involves pressing mortar into a masonry joint and forming concave grooves with a steel tool that has a concave shape. It enhances the appearance of the mortar joint but is prone to quicker damage.

Struck Pointing

Struck pointing begins by filling and pressing mortar to match the brick masonry face, followed by pressing the top edge of the joint inside around 10 mm compared to the bottom corner. This forms a slope from top to bottom for efficient rainwater drainage.



Types of Pointing

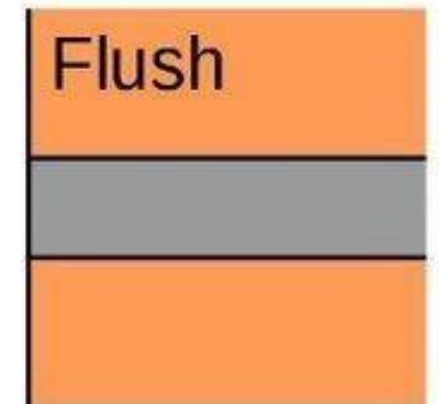
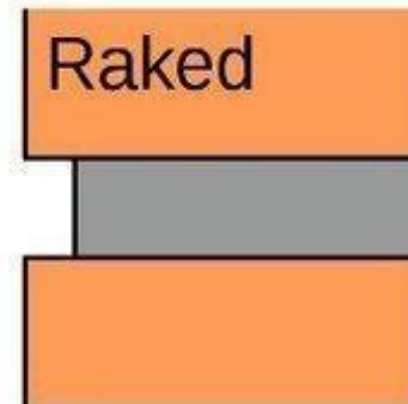
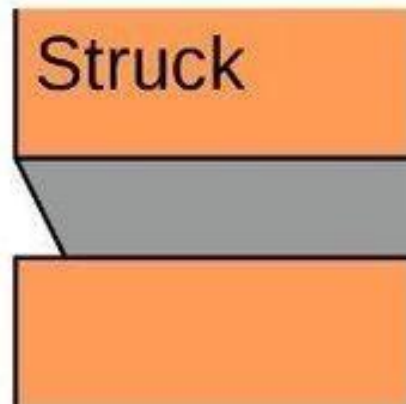
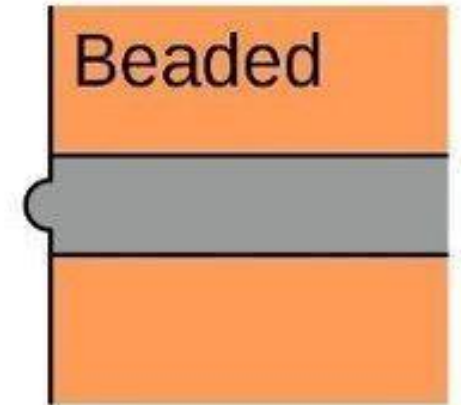
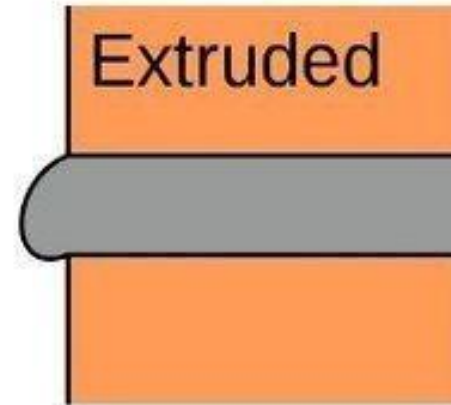
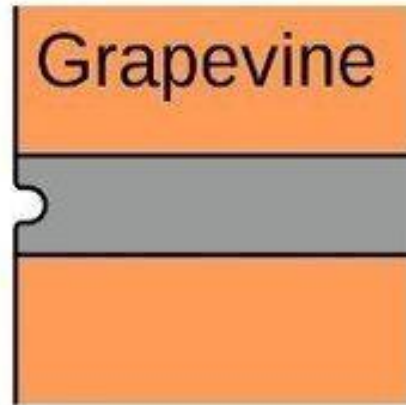
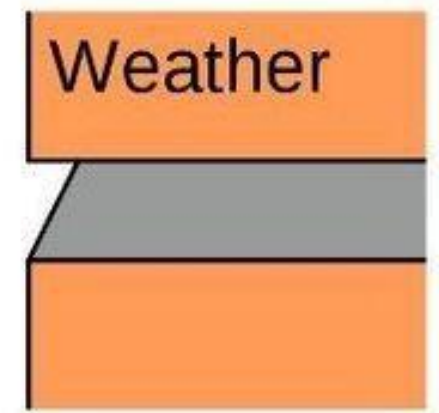
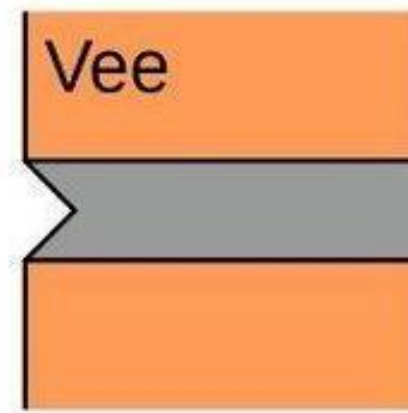
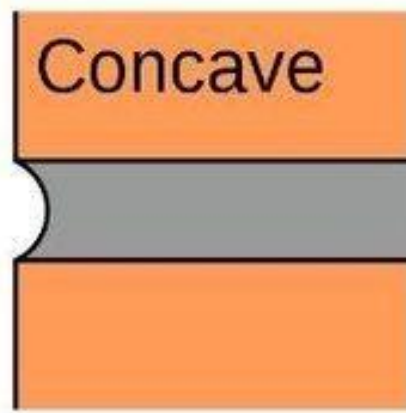
Recessed Pointing

Recessed pointing, while not suitable for exposed buildings, entails pressing mortar back from edges by 5 mm or more, with the pointed face kept vertical. It's advisable to use frost-resistant bricks for recessed joints.

Tuck Pointing

Tuck pointing involves filling and pressing the mortar joint to the masonry's surface level and then, while the mortar is still fresh, cutting grooves in the joint. These grooves are 5 mm wide and 3 mm deep and are filled with white cement putty, projecting 3 mm beyond the joint's face. When these projections are made with mortar, it's referred to as bastard pointing or half tuckpointing.

Types of Pointing





Work Process of Pointing

Pointing is carried out in a sequence of steps to ensure strong, durable and well-finished masonry joints. Proper execution of each step improves both the appearance and weather resistance of the masonry work.

The work process for Pointing includes the following steps:

- **Preparation:** Ensure the masonry surface is clean and free from debris.
- **Mix Mortar:** Prepare mortar with the right consistency.
- **Application:** Fill the joints with mortar using appropriate tools.
- **Finishing:** Shape and level the mortar in the joints to the desired profile.
- **Curing:** Allow the pointing to cure for optimal strength.
- **Cleaning:** Clean excess mortar from the masonry surface.
- **Protection:** Protect the pointing from excessive moisture during the initial curing phase.
- **Inspection:** Check for imperfections and make necessary touch-ups.
- **Final Finish:** Achieve the desired texture or appearance.



Paint - Introduction

- ❖ Paint is a liquid or semi-liquid coating material applied to a surface that dries or cures to form a continuous protective and decorative film.
- ❖ It is commonly used on concrete, cement plaster, brickwork, steel, timber, masonry, and previously painted surfaces. In construction, paint serves three major purposes: protection, decoration, and durability.
- ❖ It protects surfaces from moisture, weathering, corrosion, and other environmental effects while improving their appearance through colour, texture, and finish. When properly selected and applied, paint also helps increase the service life of construction materials and reduces maintenance requirements.



Basic Components of Paint

Paint = Binder + Pigment + Solvent + Extender + Additives

Component	Main function
Binder/Vehicle	Forms the film and binds pigment
Pigment	Provides colour and opacity
Solvent/Thinner	Controls viscosity
Extender	Improves bulk and economy
Additives	Improve specific properties

CLASSIFICATION OF PAINT: TYPES, APPLICATIONS & FUNCTIONS

COMPOSITION & BINDER TYPES

OIL PAINT

- Slow drying, durable

EMULSION PAINT

- Low odor
- Wall icon

ACRYLIC PAINT

- Fast drying, vibrant color

EPOXY PAINT

- Strong bond
- Industrial floor

POLYURETHANE PAINT

- Gloss
- Tough topcoat

CEMENT PAINT

- Rough texture
- Building icon

SURFACE / APPLICATION

MASONRY PAINT

Paints surface wall, roller and roller

METAL PAINT

Steel structure, steel structure, gear, chain

WOOD PAINT

Timber planks, brush rool and brush

FLOOR COATING

Warehouse floor, footprints, more

ROAD MARKING PAINT

Asphalt road

SPECIAL FUNCTION

ANTI-CORROSIVE PAINT

Rusty metal beam getting protected

FIRE-RETARDANT PAINT

Fire of with Fire Shield

WATERPROOF COATING

Umatter droplets running off a wall

HEAT-RESISTANT PAINT

Engine high temperature

Note: Local regulations on paint may vary



Types of Paint

Distemper Paint

- Among the most budget-friendly painting options available in India today.
- Often referred to as a form of "whitewash" due to its composition and finish.
- Primarily made from chalk, lime, water, and occasionally tinted with color pigments.
- Classified as a water-based paint, making it easy to apply and environmentally friendly.





Types of Paint

Emulsions

An emulsion is a liquid containing finely dispersed particles, often referred to as plastic paint. This water-based coating delivers a smooth, matte finish ideal for interior walls. What sets emulsion paints apart is their acrylic composition, which ensures outstanding durability and maintains a freshly painted appearance over time. These paints are washable, allowing most stains to be easily wiped away with a cloth dampened in mild soapy water. Additionally, the paint film resists fungus and mildew growth.

Coverage per coat on new surfaces:

- Exterior: 5.5 to 6 m² per litre
- Interior: 17 m² per litre





Types of Paint

Enamel Paint

Enamel paint refers to a type of coating that dries by exposure to air, resulting in a tough, often shiny surface. It's commonly applied to exterior surfaces or areas that endure heavy use and fluctuating temperatures.

Traditionally, "enamel paint" has described oil-based paints known for their glossy finish. However, in recent times, many water-based or latex paints have also adopted this label. Nowadays, the term broadly signifies paints that form a hard, durable surface, often associated with premium brands, high-gloss floor coatings, or spray paints.





Types of Paint

Varnish

- Varnish is a transparent or translucent protective coating.
- It is mainly applied to timber and wooden surfaces.
- It forms a hard, smooth and protective film after drying.
- Unlike paint, it generally does not hide the natural grain and texture of wood.

Functions

- Protects wood from moisture and dust.
- Improves resistance to wear and abrasion.
- Enhances the natural colour and grain of timber.
- Provides a smooth and attractive finish.
- Increases the durability and service life of wooden surfaces.





Defects in Painting

- **Blistering** – Formation of bubbles or raised areas in the paint film.
- **Peeling** – Paint film separates from the underlying surface.
- **Flaking** – Paint breaks into small pieces and detaches from the surface.
- **Cracking** – Fine or deep cracks develop in the paint film.
- **Chalking** – Formation of a loose, powdery layer on the painted surface.
- **Fading** – Loss of original colour due to sunlight, weathering or ageing.
- **Sagging/Running** – Wet paint flows downward, producing uneven streaks.
- **Wrinkling** – Paint film develops wrinkles during drying.
- **Efflorescence** – White salt deposits appear on masonry surfaces.
- **Brush Marks** – Visible lines or marks remain after brush application.
- **Uneven Finish** – Variations in colour, gloss or texture occur over the surface.

THANK

YOU